

Chapter 11: Probability

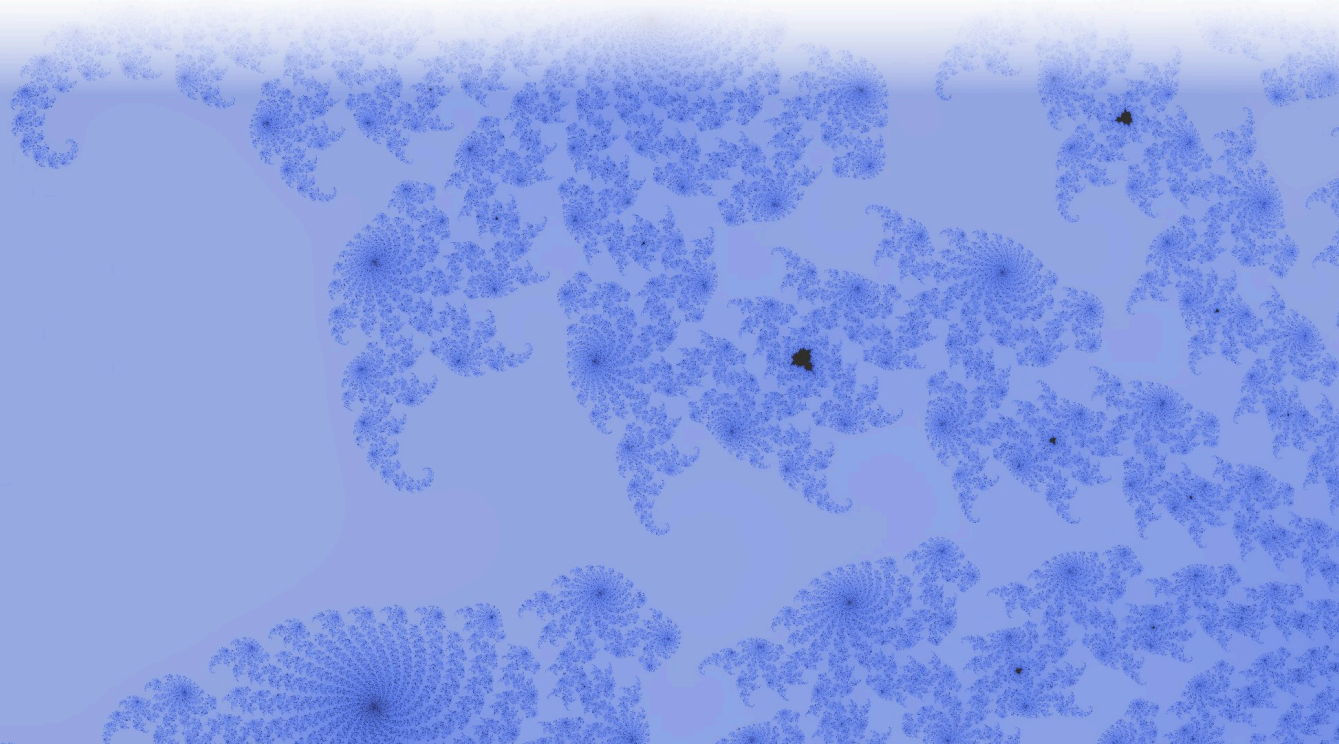
Drill practice and mixed revision of all 5 lessons · 44 questions

NSW syllabus outcome

Stage 4 Statistics and probability: MA4-PRO-C-01 solves problems involving the probabilities of simple chance experiments.

CHAPTER

11



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 11. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–31

11.01 Sample space

Easy How many outcomes are in the sample space?

- ① choosing a letter from MATHS ② choosing a digit (0 to 9) ③ choosing a vowel

Medium How many outcomes are there for both together? (Multiply.)

- ④ spin a 4-sector spinner and toss a coin ⑤ roll a die and spin a 3-sector spinner ⑥ choose an entrée (2), main (5) and dessert (3)

11.02 Probability

Easy Find the probability as a fraction in simplest form.

- ⑦ 8 yellow, 5 black: P(yellow) ⑧ 4 red, 7 blue: P(red) ⑨ 7 green, 8 red: P(green)

Medium A letter is chosen at random from the word. Find the probability.

- ⑩ PROBABILITY: P(P) ⑪ KINGSLIFF: P(I) ⑫ PROBABILITY: P(B)

11.03 The probability scale

Easy Write the chance word: impossible, unlikely, even chance, likely or certain.

- ⑬ $\frac{1}{20}$ ⑭ 95% ⑮ 0.05 ⑯ 90%

Medium Which is more likely? Change to decimals to compare.

- ⑰ $\frac{2}{3}$ or 90% ⑱ $\frac{3}{5}$ or 85% ⑲ $\frac{2}{3}$ or 30% ⑳ $\frac{1}{3}$ or 90%

11.04 Experimental probability

Easy Find the expected number: probability $\times$ number of trials.

- ⑴ an even number on a die in 12 trials ⑵ heads in 52 trials ⑶ a 1 or 2 on a die in 24 trials

Medium Find the experimental probability as a percentage. Compare it with the theoretical probability.

- ⑷ a 6 came up 6 times in 20 rolls ⑸ a 6 came up 5 times in 50 rolls

11.05 Complementary events

Easy Find P(not E): subtract P(E) from 1.

- ⑹ $P(E) = \frac{1}{6}$ ⑺ $P(E) = \frac{5}{6}$ ⑻ $P(E) = \frac{1}{9}$ ⑼ $P(E) = \frac{5}{8}$

Medium Describe the complementary event.

- ⑽ rolling less than 3 ⑾ winning

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 32–41

32 **Easy**

How many outcomes are in the sample space?
choosing a day of the week Lesson 11.01

34 **Easy**

Write the chance word for this probability.
0.95 Lesson 11.03

36 **Medium** How many outcomes are there when
rolling a die? Lesson 11.01

38 **Medium** Which word describes $P = 0.9$? Lesson 11.03

40 **Challenging** Are the outcomes of a test (pass/fail)
equally likely? Explain. Lesson 11.01

33 **Easy**

Find the expected number.
heads in 90 tosses Lesson 11.04

35 **Easy**

A bag has 3 red, 5 blue and 2 green counters. Find the
probability.
 $P(\text{blue or green})$ Lesson 11.02

37 **Medium** How many 3s are expected in 120 rolls?
Lesson 11.04

39 **Medium**
Find each probability as a fraction and a percentage (1 d.p.).
15 dark, 13 milk, 12 white chocolates: $P(\text{white})$
Lesson 11.02

41 **Challenging** A 3 came up 25 times in 120 rolls. Find
the observed $P(3)$. Lesson 11.04

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 42–44

42

Challenging

Problems.

A bag has 12 counters and $P(\text{green}) = \frac{1}{4}$. How many are green? Lesson 11.02

43

Challenging

Problems.

Explain why $P(E) + P(\text{not } E) = 1$. Lesson 11.05

44

Challenging

Mark each event on the scale.

Describe an event that is: (a) impossible (b) unlikely (c) certain. Lesson 11.03